RESULT 1 / NOTHING IS LOST

First, nothing is lost: RIGOR keeps the schema.

Every LLM, every database. We climb from the easiest check to the hardest.

1 VALID | 2 KEEPS THE SCHEMA | 3 ADDS MEANING | 4 FEWER ERRORS | 5 GOLD MAPPINGS | 6 JUDGES

1 VALID

30 / 30

ontologies are valid OWL 2 DL and logically consistent, for every method and LLM.

2 KEEPS THE SCHEMA

share of database columns covered, per ontology (%)

	Liver Cancer			eICU-CRD			Chinook			Avg
	Claude	Mistral	DeepSeek	Claude	Mistral	DeepSeek	Claude	Mistral	DeepSeek	
Direct Mapping	97			99			78			**91%**
LLM baseline	94	31	89	96	68	86	100	50	52	**74%**
Non-Iterative	41	6	12	22	28	46	58	16	78	**34%**
RIGOR	**91**	**99**	**87**	**100**	**99**	**77**	**95**	**95**	**95**	**93%**

Without the seed, coverage swings with the LLM: **Non-Iterative 6–78%.** **With RIGOR, every LLM keeps ≥ 77% in every database.**

Validity: RDFLib; consistency: HermiT. Coverage: a column counts as covered when an ontology class or property name is similar to it (embedding similarity ≥ 0.55). Paper Sec. 4.6–4.7, Table 9.

RESULT 2 / MEANING AND ERRORS

Then, meaning is added, with far fewer errors.

What each method adds, and the modeling errors per ontology, for every LLM and database.

1 VALID | 2 KEEPS THE SCHEMA | 3 ADDS MEANING | 4 FEWER ERRORS | 5 GOLD MAPPINGS | 6 JUDGES

3 ADDS MEANING

elements per ontology (avg)

	Labels	Provenance	Hierarchy
Direct Mapping	0	0	0
LLM baseline	0	0	0
Non-Iterative	**389**	0	0
RIGOR	**777**	**196**	**59**

Same pattern for every LLM and database.

4 FEWER ERRORS

OOPS! modeling pitfalls per ontology

	Liver Cancer			eICU-CRD			Chinook			Avg
	Claude	Mistral	DeepSeek	Claude	Mistral	DeepSeek	Claude	Mistral	DeepSeek	
Direct Mapping	731			665			102			**499**
LLM baseline	422	397	403	434	440	421	95	74	81	**307**
Non-Iterative	29	61	18	39	220	27	24	35	14	**52**
RIGOR	**20**	**23**	**3**	**2**	**2**	**26**	**0**	**0**	**0**	**8**

Without validation, the smaller LLM fails hardest: **Mistral on eICU, 220.** With RIGOR: **2, and zero on Chinook for every LLM.**

Paper Table 8 (annotations, prov:wasDerivedFrom, SubClassOf) and Table 7 (OOPS! totals). 72% of the other methods' pitfalls are missing annotations (P08).

RESULT 3 / EXTERNAL BENCHMARK

RIGOR keeps the link to the data.

BURR [Laskowski et al., PACMMOD 2025] · ISWC scenario · highest F1 among ontology-learning systems

1 VALID | 2 KEEPS THE SCHEMA | 3 ADDS MEANING | 4 FEWER ERRORS | 5 GOLD MAPPINGS | 6 JUDGES

WHAT BURR CHECKS

Every ontology element must map back to the database.

CLASSES	`person`	→	`Class: Person`
RELATIONS	`paper.author_id`	→	`ObjectProperty: author`
ATTRIBUTES	`person.name`	→	`DataProperty: name`

Elements match when they map to the same database elements, whatever their names. F1 per element type.

Illustrative elements · ISWC gold mapping: D2RQ example by Bizer & Seaborne

SYSTEM	CLASSES	RELATIONS	ATTRIBUTES
W3C-Mapper / Ontop rule-based	0.53	0.11	0.39
RDB2Onto rule-based	0.67	0.62	0.44
OntoGenix LLM-based	0.58	0.35	0.44
RIGOR LLM + rules	**0.73**	**0.67**	**0.52**

Relations: 0.67 vs 0.11 for W3C-Mapper / Ontop. Foreign keys become named relations, like hasPatientData.

One scenario · 9 tables · 46 columns · 11 FKs · RIGOR with Claude · its prov:wasDerivedFrom links exported as a D2RQ mapping · Source: paper Table 13

RESULT 4 / LLM JUDGE AND HUMAN EXPERTS

An LLM judge and human experts both rank RIGOR first.

Blind rankings of four anonymized ontologies per table, broken down by database.

1 VALID | 2 KEEPS THE SCHEMA | 3 ADDS MEANING | 4 FEWER ERRORS | 5 GOLD MAPPINGS | 6 JUDGES

LLM-AS-A-JUDGE · GPT-5.4

58 / 58 tables with RIGOR ranked first

average score, 0–5 (Claude ontologies)

	Liver Cancer	eICU-CRD	Chinook
Direct Mapping	2.05	2.13	0.13
LLM baseline	1.38	0.28	1.48
Non-Iterative	1.89	0.67	2.77
RIGOR	**3.93**	**3.64**	**3.32**

10 ONTOLOGY EXPERTS

28 / 40 rankings with RIGOR first (70%)

first places, of 20 per database

	Liver Cancer	eICU-CRD
Direct Mapping	3	2
LLM baseline	1	4
Non-Iterative	2	0
RIGOR	**14**	**14**

Direct Mapping: structure without meaning. LLMs alone: meaning without structure. **RIGOR: both, and checked.**

Paper Table 1 (scores, rankings; judge: 58 tables, 290 competency questions) and Table 12 (expert agreement, Kendall's W 0.39–0.72; 4 medical tables).

RESULT 5 / WHERE THE GAINS COME FROM

RIGOR's gains do not depend on model size or memorized data.

Is it just a strong LLM? Is it memorized data? We broke the results down by LLM and by domain.

1 VALID | 2 KEEPS THE SCHEMA | 3 ADDS MEANING | 4 FEWER ERRORS | 5 GOLD MAPPINGS | 6 JUDGES

ACROSS LLMs · MISTRAL SMALL (24B)

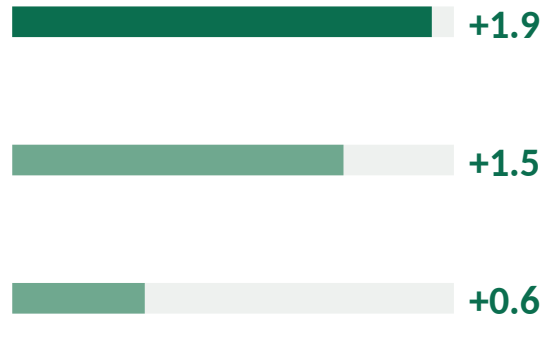
column coverage, average over the three databases



On par with Claude under RIGOR (**95%**), for **$0.12** instead of $33 for all three databases.

ACROSS DOMAINS · LLM JUDGE

RIGOR's lead over the best other method (score 0–5)



A small open model catches up, and the lead grows where the LLM knows least: **grounding and validation make the difference.**

Coverage: paper Table 9 (Claude 95%, DeepSeek 86% with RIGOR). Cost: Table 15. Judge scores: Table 1 (Claude ontologies; best other: Direct Mapping, Direct Mapping, Non-Iterative).